

Session 1: AI for Coding – Tools, Concepts, and Setup

Elira Kuka & Tanner Regan

The George Washington University

What We Will Cover (Sessions 1 & 2)

- ▶ Part 1: Tools, concepts, and setup
 - Overview of AI coding tools and how to choose among them
 - How to use AI Agents: Terminal, Desktop App, VS Code
 - Conceptual overview of context engineering with Claude
 - VS Code and Claude Code integration
 - Git and Github: why they matter for research and Claude

What We Will Cover (Sessions 1 & 2)

- ▶ Part 1: Tools, concepts, and setup
 - Overview of AI coding tools and how to choose among them
 - How to use AI Agents: Terminal, Desktop App, VS Code
 - Conceptual overview of context engineering with Claude
 - VS Code and Claude Code integration
 - Git and Github: why they matter for research and Claude
- ▶ Part 2: Practical workflows
 - Writing, debugging, and automating code with AI
 - “Vibe coding” from plain English descriptions
 - Project organization and replication packages
 - Demo: data cleaning script from scratch (python and stata)

Coding Perhaps Highest-Value Use of AI

- ▶ AI most reliable when output is verifiable: code runs or it does not
- ▶ Economists spend enormous time on repetitive coding tasks
 - AI can handle most of them
- ▶ Massive productivity gains across skill levels:
 - Beginners: “vibe code” to write code in an unfamiliar language (no syntax!)
 - Advanced users: automate drudge work, prototype faster, reduce bugs

Coding Perhaps Highest-Value Use of AI

- ▶ AI most reliable when output is verifiable: code runs or it does not
 - ▶ Economists spend enormous time on repetitive coding tasks
 - AI can handle most of them
 - ▶ Massive productivity gains across skill levels:
 - Beginners: “vibe code” to write code in an unfamiliar language (no syntax!)
 - Advanced users: automate drudge work, prototype faster, reduce bugs
- ⇒ First two sessions will cover how to use AI to complete coding tasks

Overview of AI Coding Tools

The AI Coding Landscape: Many Options

- ▶ General-purpose AI assistants (good at coding, among many other things):
 - **Claude** (Anthropic) — our focus today
 - **ChatGPT / GPT-4o** (OpenAI)
 - **Gemini** (Google)
- ▶ Agentic coding tools (read files, write and run code autonomously):
 - **Claude Code** (Anthropic) — our focus today
 - **Codex** (OpenAI) — closest competitor to Claude Code
 - **GitHub Copilot** (Microsoft) — inline suggestions in editor
 - **Cursor / Windsurf** — AI-first code editors
- ▶ How to choose: task type, privacy needs, and workflow integration

Chatbot vs. Agentic Tool

- ▶ **Chatbot** (Claude.ai, ChatGPT): you ask → it answers → *you* act
- ▶ **Agentic tool** (Claude Code): you give a goal → it reads files, writes code, runs it, fixes errors, iterates → reports back
- ▶ For economists: instead of “here is some code,” it runs the code and tells you what it found

Which One Should You Use? Claude Code vs. Codex

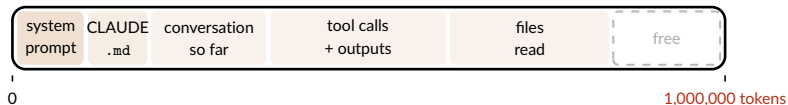
- ▶ Two serious options for agentic coding today: **Claude Code** and **Codex**
- ▶ Plans mirror each other at each price point:
 - \$20/mo: Claude Pro vs. ChatGPT Plus
 - \$100/mo: Claude Max (5x) vs. ChatGPT Pro (5x)
 - \$200/mo: Claude Max (20x) vs. ChatGPT Pro (20x)
- ▶ Both use a 5-hour rolling window + weekly cap; limits reset automatically
- ▶ At the same price, Codex currently gives more usage per dollar

Which One Should You Use? Claude Code vs. Codex

- ▶ Two serious options for agentic coding today: **Claude Code** and **Codex**
- ▶ **Codex** has some advantages:
 - More usage per subscription dollar — important for heavy users
 - Sometimes stronger on pure coding benchmarks
- ▶ **Claude Code** has some advantages:
 - Often better at tasks beyond code: understanding context, writing, reasoning
- ▶ **Bottom line:** this comparison is fast-moving and will keep changing
 - I am using Claude Code and find it excellent for my workflow
 - I would switch only if Codex proved clearly and consistently superior
 - My advice: pick one, learn it well, reassess in 6 months

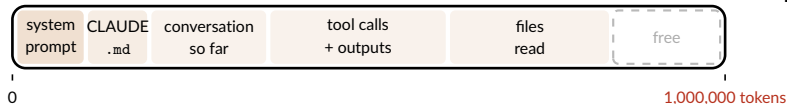
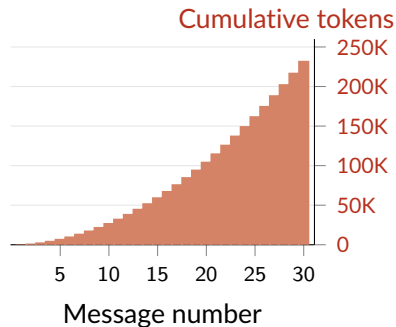
What You Pay For: Tokens and Context

- ▶ **Token:** the unit AI reads and writes — roughly $\frac{3}{4}$ of a word
 - **You pay for:** tokens *in* (prompt + files) and tokens *out* (reply) — input is cheaper
- ▶ **Context window:** everything the model holds at once — fills as session grows



What You Pay For: Tokens and Context

- ▶ **Token:** the unit AI reads and writes — roughly $\frac{3}{4}$ of a word
 - **You pay for:** tokens *in* (prompt + files) and tokens *out* (reply) — input is cheaper
- ▶ **Context window:** everything the model holds at once — fills as session grows



Getting the Most Out of Your Subscription

- ▶ Subscriptions have rolling limits — fewer tokens per task means more use
- ▶ How to use fewer tokens:
 - Start a new session for each new task: context accumulates and raises costs
 - Store standing instructions once so you never repeat yourself
 - Be specific: vague prompts waste tokens and invite unfocused replies
 - Ask multiple things at once so less back and forth
 - Compact your chats/context window
- ▶ Silly trick: start Claude when wake up so 5-hour window ends earlier in day

Claude Code / Codex Interfaces

Three Ways to Use an Agentic Coding Tool

▶ **Terminal** (command line):

- Most powerful; works directly in your project directory
- Best for: running scripts, data analysis, replication tasks

▶ **Desktop App:**

- Friendlier interface; easier to get started
- Projects feature: persistent context across sessions
- Best for: drafting, quick questions, early exploration

▶ **VS Code integration:**

- Agent lives inside your editor — see code and AI side by side
- Best for: sustained development work, Git integration, Jupyter/Python

Terminal vs. VS Code: Quick Comparison

Terminal (CLI)

- ▶ Lightweight; fast startup
- ▶ Works anywhere, incl. remote servers
- ▶ Keyboard-driven; fast for quick tasks
- ▶ No inline diffs; steeper setup

VS Code Extension

- ▶ Inline diffs; accept changes block-by-block
- ▶ Code & AI panel side by side
- ▶ Best for large codebases, sustained work
- ▶ Heavier on CPU and memory

Live Demo: Terminal

- ▶ ▷ *Demo slide – minimal text, walk through live*
- ▶ Navigate to a project directory
- ▶ Ask Claude Code to describe what it sees
- ▶ Ask it to write a simple data task

Live Demo: Desktop App

- ▶ ▷ *Demo slide – minimal text, walk through live*
- ▶ Create a Project and load context
- ▶ Same task as terminal – compare experience

Live Demo: VS Code

- ▶ ▷ *Demo slide – walk through VS Code interface live*
- ▶ Same task as terminal – compare experience

Contex Engineering with Claude Overview

Context Engineering

▶ Context Window

- ▶ Each prompt to Claude Code bundles the entire history of your session
 - ▶ your prompts, Claude's responses, files it read, code outputs, system prompts
- ▶ These are all appended until the context window hits its fixed size limit
 - ▶ For Claude Code this is around 200k tokens
- ▶ Claude then “**auto-compacts**” writes its own summary and then restarts fresh
 - ▶ less preferable to frequent intentional compaction

▶ Context Engineering

- ▶ You can tell Claude when to compact using `/compact your instructions`
- ▶ Develop habits that the context window focussed on relevant information
 - ▶ Claude.md vs skills —and— ‘ask before edits’ vs ‘plan’ Modes

Read more at [Paul GP's Guide](#)

Anatomy of a context window (example from Paul GP's Guide)

```
<|start|>system<|message|>You are ChatGPT, a large language model trained by OpenAI.  
Knowledge cutoff: 2024-06  
Current date: 2025-06-28  
Reasoning: high  
# Valid channels: analysis, commentary, final.  
Channel must be included for every message.  
Calls to these tools must go to the commentary channel: 'functions'.<|end|>
```

system prompt
(written by openAI)

ANATOMY OF A
MODERN LLM
CHAT

```
<|start|>developer<|message|># Instructions  
Be concise and technical. Provide code examples when relevant.
```

developer
message
(tool definitions)

```
# Tools  
## functions  
namespace functions {  
  type web_search = (_: {  
    query: string,  
    num_results?: number,  
  }) => any;  
} // namespace functions<|end|>
```

user
prompt

```
<|start|>user<|message|>What's new in the latest version of polars?<|end|>
```

thinking

```
<|start|>assistant<|channel|>analysis<|message|>User asking about latest polars version.  
My knowledge cutoff is 2024-06, need to search for recent updates.<|end|>
```

```
<|start|>assistant<|channel|>commentary<|message|><|call|>functions.web_search  
{  
  "query": "polars python latest version new features", "  
  "num_results": 5  
}<|end|>
```

tool call

```
<|start|>tool<|message|>[{"title": "Polars 1.8.0 Release Notes", "snippet": "New  
streaming engine, improved join performance, added pivot_wider...", "url": "..."}]<|end|>
```

tool
results

```
<|start|>assistant<|channel|>final<|message|>Polars 1.8.0 (latest as of Nov 2025)  
includes:
```

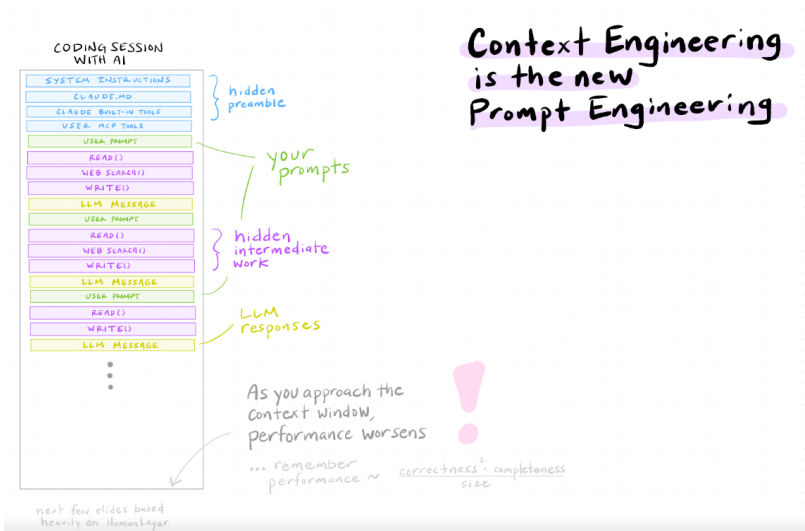
final response

```
- **Streaming engine overhaul**: Better memory efficiency for large datasets  
- **Join performance**: 2-3x faster for certain join types  
- **New API**: `pivot_wider()` for reshaping data  
- **LazyFrame improvements**: Enhanced query optimization
```

```
Check the release notes for migration details if upgrading from older versions.<|return|>
```

You
typically
see only
these
two!

Context engineering (example from Paul GP's Guide)



Context auto-compact (example from Paul GP's Guide)

If you hit the
content window
the LLM will
auto-compact



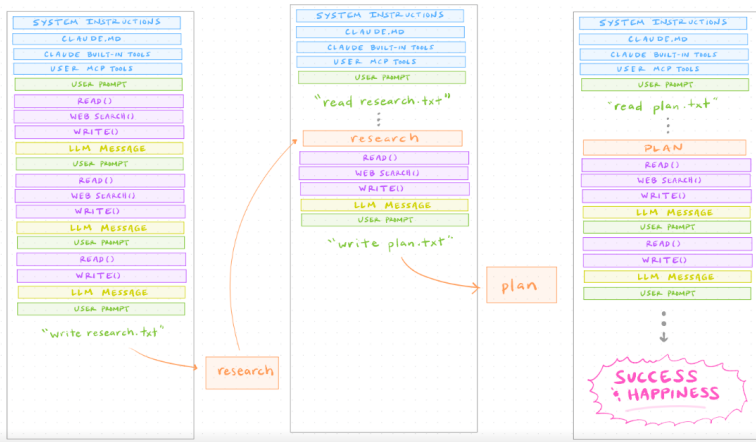
the model summarizes your conversation

...this is
rarely
ideal...



Context intentional compacting (example from Paul GP's Guide)

Frequent intentional compaction is better



CLAUDE.md and Skills

- ▶ **CLAUDE.md** (memory) — Claude reads at the start of every session
 - Store context that is always true so you don't repeat yourself in every prompt
 - `.claude/CLAUDE.md` is global, `your-project/CLAUDE.md` is *project* specific
 - Ask Claude to update after substantial changes to project
- ▶ **Skills** — Claude reads every time it is called
 - Store context for a task that you repeat often, but not always relevant
 - `.claude/skills/your-skill/SKILL.md` is called with `/your-skill`
 - Write and update your own skills or download from others

Read more at [Claes Backman's Guide](#) (full descriptions, examples of CLAUDE.md and skills, download pre-made skills)

your-project/CLAUDE.md (example from Claes Backman's Guide)

```
# Project: [Paper title]

## Data
- Unit of analysis: firm-year panel, 2010-2022
- Raw data lives in data/raw/ – never edit these files
- Main dataset after cleaning: data/clean/panel.dta (N ≈ 47,000)

## Code conventions
- R scripts are numbered (01_clean.R, 02_analysis.R, ...)
- All regressions cluster SEs at the firm level
- Use fixest for all panel regressions

## LaTeX
- Main file: paper/main.tex
- Tables generated by 03_tables.R go to paper/tables/
- Use \widehat{} not \hat{} for estimators
```

`.claude/skills/robustness/SKILL.md` (example from Claes Backman's Guide)

```
---  
name: robustness  
description: Run standard robustness checks on the main regression.  
---
```

For the main specification in this project:

1. Re-run with alternative clustering (industry-year instead of firm)
2. Re-run dropping the top and bottom 1% of the outcome variable
3. Re-run on the pre-2020 subsample only
4. Produce a summary table comparing coefficients across all specs

Claude Modes

▶ Ask before edits

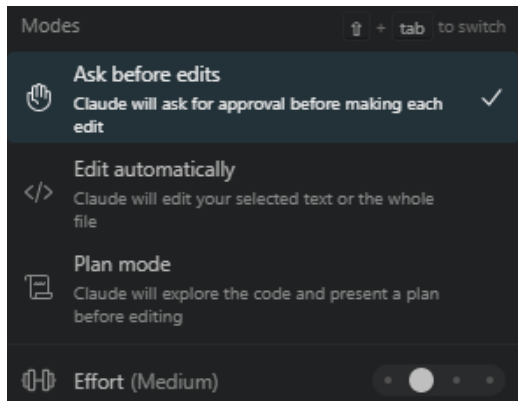
- ▶ Do steps one at a time
- ▶ Asks your permission before running each and every command

▶ Edit automatically

- ▶ Same as above but runs commands without asking permission (risky)

▶ Plan

- ▶ Claude writes plan.md with many steps
- ▶ You review and execute entire plan



Other tips for managing the context window

- ▶ **Use @-mentions to be selective**

- ▶ Rather than Claude reading everything, point to a specific file

- ▶ e.g. `update all the pathnames in @Lecture_2.tex`

- ▶ **Check usage with /context**

- ▶ Type `/context` in the panel to see a breakdown of how your context window is being used and how much space remains.

VS Code with Claude Overview

What is VS Code?

- ▶ Integrated Development Environment (IDE)
 - ▶ Free, open source (created by Microsoft), and very popular
- ▶ Core features:
 - **Explorer:** file browser and project tree
 - **Editor:** syntax highlighting for every language (Stata, R, Python, $\text{\LaTeX}$, ...)
 - **Integrated terminal:** run scripts without leaving the editor
 - **Extensions:** Git, Claude Code, Jupyter, and more
 - **Command Palette** (`Ctrl+Shift+P`): search and run any action by name
- ▶ Claes Backman's Guide [Claude Code in VS Code for Academic Economists](#)
 - ▶ Other resources: [Official intro videos \(Microsoft\)](#), [Tips and Tricks \(Microsoft\)](#)


```
File Edit Selection View Go Run Terminal Help
Lecture_1.tex 7, M x preamble.tex
les > Lecture_1.tex > [textcolor{violet}{VS Code with Claude Overview \vspace{0.3in}}] > [1] Frame: VS Code with Claude Overview
375 \section{\textcolor{violet}{VS Code with Claude Overview \vspace{0.3in}}}
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863
864
865
866
867
868
869
870
871
872
873
874
875
876
877
878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917
918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971
972
973
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
```

Lecture_1.pdf M x

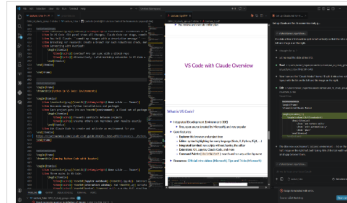
GWU_AI_study_group > slides > Lecture_1.pdf

27/40

What is VS Code?

- Integrated Development Environment (IDE)
 - Free, open source (created by Microsoft), and very popular
- Core features:
 - Explorer: file browser and project tree
 - Editor: syntax highlighting for every language (Stata, R, Python, etc.)
 - Integrated terminal: run scripts without leaving the editor
 - Extensions: Git, Jupyter, Claude Code, and more
 - Command Palette (Ctrl+Shift+P): search and run any action by name
- Resources: Official intro videos (Microsoft), Tips and Tricks (Microsoft)

28/40



VS Code with Claude Overview

Set up Claude.md for AI economics study g...

Set up Claude.md for AI economics study g...

c:\Users\tanner_regan\Docu...

the slide at line 510 should be split in half vertically so that the list is on the left and image is on the right.

- Thought for 1s >
- Let me read the slide at line 510.
- Read c:\Users\tanner_regan\Documents\GitHub\GWU_AI_study_group\slides\Lecture_1.tex (lines 501-540)
- Now I can see the "Claude Modes" frame. I'll split it into a two-column layout with the list on the left and the image on the right.
- Edit c:\Users\tanner_regan\Documents\GitHub\GWU_AI_study_group\slides\Lecture_1.tex

Added 4 lines

```
~~~~~
\begin{frame}
\frametitle{Claude Modes}

\begin{columns}[T]
\begin{column}[0.48\textwidth]
\begin{itemize}
\item 'ask before edits'
\item 'edit automatically'
\item 'plan'
\end{itemize}
\end{column}
\end{columns}
\end{frame}
```

The slide now uses Beamer's `columns` environment — list on the left half, image on the right half, both taking 48% of the text width with a small gap between them.

c:\Users\tanner_regan\Docu...

ctrl esc to focus or unfocus Claude

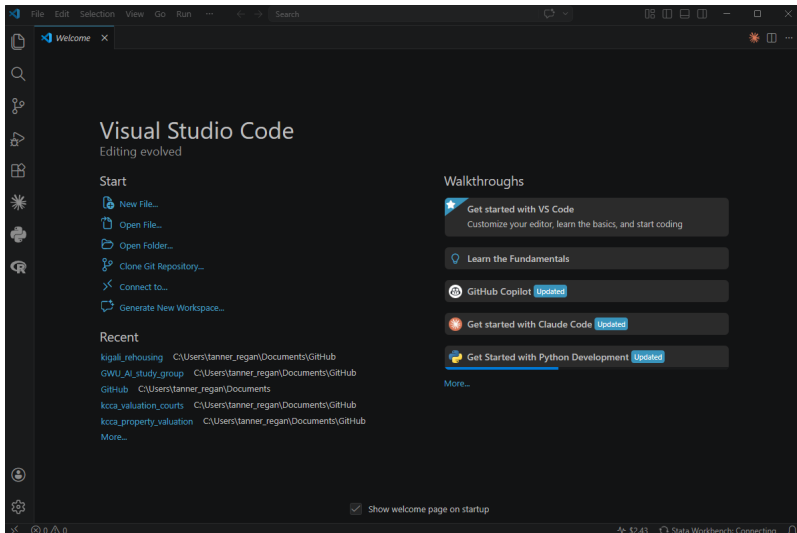
+ Lecture_1.tex Ask before edits

Filter (e.g. text, "text", "text")

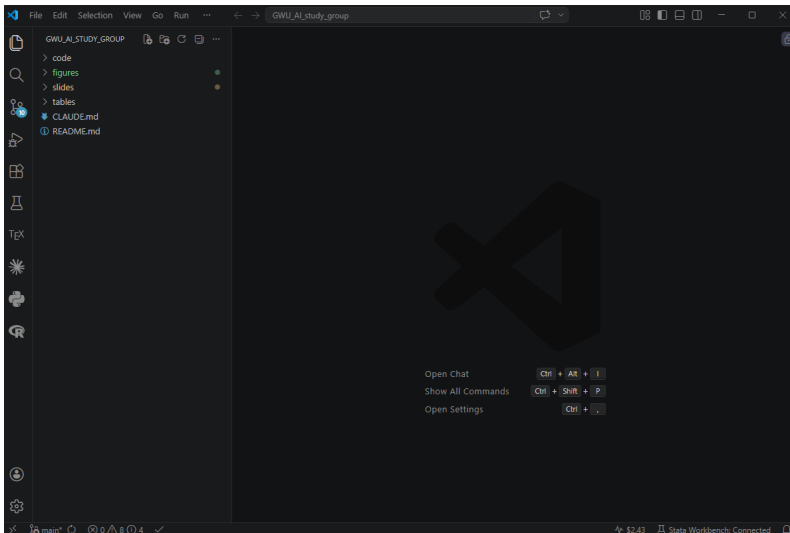
Claude Code **extension** in VS Code

- ▶ Give Claude access to your whole project directory to maximize its usefulness
 - ▶ Open your full project folder in VS Code (File → Open Folder)
 - ▶ This can be any folder on your computer — plain folder, a git repository, a Dropbox folder, etc.

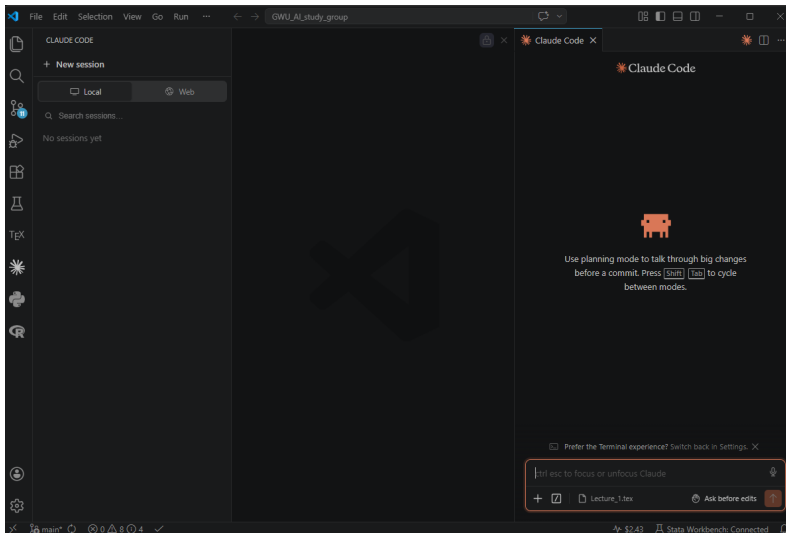
New VS Code Window — open new folder



Project folder is open — now open a new Claude Code session

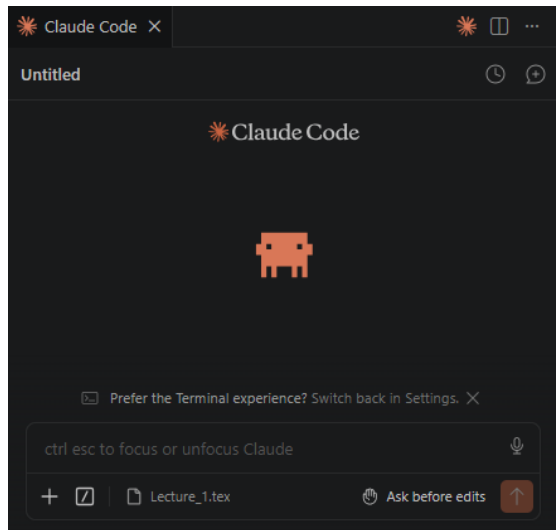


New Claude Code session is open



Claude Code extension in VS Code

- ▶ Each new conversation in the Claude panel is a fresh context
 - ▶ Click the + button at the top of the Claude panel to start one,
 - ▶ or type `/clear` to reset
- ▶ Go over a fresh window →
 1. Session Title
 2. Session History
 3. Mode menu
 4. Current File Selection
 5. Attachments
 6. Command Menu



Files and Data that Claude is good at reading

- ▶ Claude Code works best with plain text files
 - ▶ e.g. `.txt`, `.do`, `.py`, `.tex`, `.csv`, `.log`, `.md`, etc.
- ▶ Tips for optimal performance with other file types
 - ▶ `.docx`: convert to markdown with `pandoc file.docx -o file.md`
 - ▶ `.xlsx`, `.dta`, `.rds`, `.sav`: convert to CSV first with a short script
 - ▶ `.pdf`:
 - ▶ Use the `.tex` file whenever possible (especially if equations and tables)
 - ▶ Long PDFs (50+ pages): convert to markdown first
 - ▶ Scanned PDFs: run OCR first, then convert to markdown
- ▶ Create a skill that loads file types, e.g. `/pdf-to-markdown`

Read more at [Claes Backman's Guide](#) (details on handling different file types)

Git: Version Control for Coding Projects

Why Git Matters for Research

- ▶ Version control = a complete history of every change to your code
 - ▶ No more: `analysis_v3_FINAL_revised2.do`
- ▶ Enables:
 - Backup: retrieve any prior version of your code
 - Collaborate: coauthors and RAs edit without overwriting each other
 - Track: what changed and who made the changes at every stage
- ▶ Can help improve project replicability both internally and externally
 - ▶ an automatic audit trail
 - ▶ sharing and iterating on projects publicly

Read more at Luís Fonseca's [Version Control with git \(for economists\)](#)

Git and Github Concepts

- ▶ Key concepts to remember
 - ▶ **Repository (repo)**: a folder whose history Git is tracking
 - ▶ **Commit**: a snapshot of the code at a point in time, with a descriptive message
 - ▶ **Push / Pull / Sync**: sync your local repo with the online/remote copy
- ▶ Other concepts, less critical to understand at the start:
 - ▶ **Branch**: a parallel version of the code for exploration
 - ▶ **Merge**: fold a branch back into the main code
 - ▶ **Clone**: copy your online repo to local computer for the first time
 - ▶ **Fork**: copy someone else's online repo to your own GitHub account

Git vs. GitHub: Two Separate Things



git — the software that runs locally

- ▶ All history lives in the `.git` folder
- ▶ Records *commits*, *branches*, *merges*
- ▶ Compares any two versions line-by-line; flags conflicts



GitHub — online hosting platform

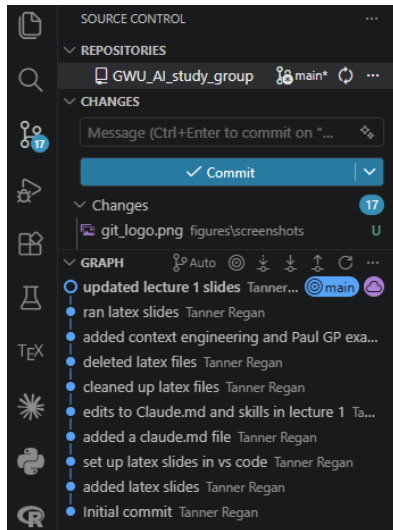
- ▶ Stores a remote copy of your repo
- ▶ Handles *Push*, *Pull*, *Clone*, *Fork*
- ▶ Free unlimited private repos
- ▶ Alternatives: GitLab, Bitbucket

Typical Git workflows for an economist

- ▶ Typical workflow for an economist:
 1. Pull/Sync code at start of work session (get latest version of online repo)
 2. Work on code; when something works, *commit* it with a descriptive message
 3. Push/Sync at end of work session (updates the online repo)
- ▶ More elaborate, with RA review
 1. RA pulls repo, creates a branch from the latest commit
 2. RA works on branch, makes commits and pushes it when finished
 3. You compare the branch to the main branch
 - ▶ recall git highlights all the differences line by line and flags conflicts
 4. You merge back into main branch, or ask RA for fixes
 5. Push/Sync to update the online repo

Git source control panel in VS Code

- ▶ Source Control panel
 - ▶ Click the github icon to open
- ▶ Go over an example →
 1. Icon with number of changes
 2. Message bar for commit
 3. Pull/Sync button
 4. commit button (also push)
 5. List of changes
 6. Graph of commits and branches



Setting up a github repository

- ▶ Go to github.com
 - ▶ Create an account, add new repository, add collaborators
 - ▶ Create a readme file that describes the project
- ▶ Open VS Code git extension (and sign into your GitHub account)
 - ▶ clone the repo to local computer: `Ctrl + Shift + P` (Windows/Linux) or `Cmd + Shift + P` (Mac) to open the Command Palette. Type `Git: Clone` and press Enter. Choose from the list of your repos.
- ▶ I have created a GitHub repo for our workshop [GWU_AI_study_group](#)
 - ▶ It is setup as Public, so you can already look at it and even Fork it
 - ▶ If you make a GitHub account and send me your username, I will add you as a collaborator so you can push changes to the repo as well

What Goes in a Repository?

- ▶ **Do NOT** store any datasets (source or generated)
 - instead store in [Box](#) (GW-endorsed secure storage)
 - or locally, Dropbox, Google Drive, etc.
- ▶ **Do** store all other project materials
 - code (Stata, Python, $\text{\LaTeX}$)
 - output: tables, figures
 - CLAUDE.md, notes, slides

```
GWU_AI_study_group/  
├── slides/  
│   ├── _preamble.tex  
│   ├── Lecture_1.tex  
│   └── Lecture_2.tex  
├── code/  
├── figures/  
├── notes/  
├── tables/  
├── CLAUDE.md  
└── README.md
```

Git and Claude Code Integration

- ▶ Claude Code is get-aware
 - ▶ When the open project is a git repo, Claude can see the entire commit history
- ▶ Claude can facilitate git operations
 - ▶ Write commit messages `commit my changes with a descriptive message`
 - ▶ Explain what changed `what changed between commit A and B?`
 - ▶ Resolve a merge conflict `fix RA code to merge with the main branch`

Read more at [Claes Backman's Guide](#)

Setting Up Your Environment

What You Need to Install

- ▶ Core tools (all free except AI subscription):
 1. **VS Code** — the editor (free)
 2. **Claude Code** — VS Code extension + CLI (requires subscription)
 3. **Git** — version control (free); create a GitHub account
 4. **Python** — Anaconda distribution recommended (free)
- ▶ VS Code extensions to install:
 - Claude Code, GitLens, Jupyter, Python, Stata Enhanced (if applicable)
- ▶ We have a step-by-step setup guide — see handout / shared folder

- ▶ What to install / setup before next class (if you want to follow along)
- ▶ VS Code, Git, Python, Claude Code subscription
- ▶ Github account
- ▶ VS Code extensions: Claude Code, Jupyter, Python, Stata

Session 1 — Key Takeaways

1. AI has perhaps most value added in coding
2. Agentic $\neq$ chatbot: reads files, writes and runs code, fixes errors, reports back
3. Two serious agentic options today: Claude Code and Codex
 - Pick one, learn it, reassess in 6 months
4. Tokens accumulate fast in long session — start new session for each new task
5. Three interfaces (Terminal, Desktop, VS Code) and Git are your essential stack

Questions?

- ▶ Questions on tools, concepts, or setup?
- ▶ Thoughts on how this fits your current workflow?